

SHH gene exploration for the ensembl ICGEB workshop

Practical Assignment

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1. Find and characterize the SHH locus

- 1) Search for SHH in the human genome in Ensembl
 - a) What is the **Ensembl Gene ID**?
 - b) On which chromosome is SHH located?
 - c) What are its genomic coordinates in the current human assembly?
 - d) On which strand is the gene located?
 - e) What is the approximate genomic span of the SHH locus?
- 2) Identify the genomic assembly being used
 - a) Is the SHH location the same between **GRCh37** and **GRCh38**?
 - b) If the coordinates differ, explain why.
- 3) Examine the **Gene** → **Region in detail** view.
 - a) What genes are located upstream and downstream of SHH?
 - b) Are there other transcripts or regulatory features surrounding the locus?

2. Explore SHH transcripts deeply

The Ensembl gene page provides a transcript table containing splice variants, transcript/protein lengths, biotypes and annotations such as MANE, APPRIS and TSL. ([Ensembl Archive documentation](#))

- 4) How many transcripts are currently associated with **SHH**?

Get for each SHH transcript, record:

Transcript	Biotype	Transcript length	Protein length	MANE?	APPRIS?	TSL

- 5) Do all SHH transcripts encode a protein?
- 6) Compare the exon structures of the different SHH transcripts. Which exons are shared?
- 7) Download the first CDS sequence for the MANE transcript
- 8) Which SHH transcript is designated as **MANE Select**?
 - a) What does MANE Select mean?
 - b) Why might using a standardized transcript be important when reporting a variant?
- 9) Compare the SHH Ensembl transcript with its corresponding **RefSeq transcript**.
 - c) Are the exon structures identical?
 - d) Are the transcript and protein sequences identical?
 - e) What are the corresponding RefSeq identifier of the main Transcript ?

3. Explore naturally occurring variation in SHH

Ensembl's Variation resources allow you to examine variants associated with genes and transcripts, population frequencies, phenotypes, LD information, and links to external resources.

- 10) If there, select one the variants you find:
- 11) What variant groups are found ?

4. Population variation

- 12) Examine the population frequency information for your SHH variant **rs104894047**.
- 13) Which populations have the highest and lowest reported allele frequencies?
- 14) Is the variant present in multiple populations or apparently restricted to particular populations?

15) What is the source of this variant ?

5. External database investigation

Ensembl provides cross-references to resources such as UniProt, RefSeq, InterPro and other databases; these links can be accessed from gene/transcript records.

16) What is the source of this variant ?

17) From the SHH Ensembl page, identify the [external database identifiers](#) for:

- a) HGNC
- b) NCBI Gene
- c) UniProt
- d) OMIM